

Patrick Meng
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Education

Rutgers University, New Brunswick, September 2023 - Present

- PhD, Computer Science, Robotics

Columbia University, New York City, September 2021 - December 2022

- Masters, Computer Science, Machine Learning Track

Rutgers University, New Brunswick, September 2017 - May 2021

- Summa Cum Laude, Honors College, Highest Honors in Computer Science, Deans List
- Magison Award for Computer Science
- Mathematics and Computer Science Majors; Statistics Minor

Research Experience

Rutgers Pracsys Lab, September 2023-Present

Advisor: Kostas Bekris

- **Robot Camera Calibration**
 - Utilized foundation models to create accurate robot camera calibrations
- **Tensegrity Control**
 - Constructed tetherless tensegrity robot, a form of soft-rigid hybrid robot, from scratch
 - Published first real-world, closed-loop experiments on tensegrity robots

Computational Biomedicine Imaging and Modeling Lab, May 2024-Present

Advisor: Dimitris Metaxas

- **Diffusion based intra-oral data generation**
 - Collaborated with Colgate-Palmolive to generate synthetic images using diffusion methods

Columbia Roam Lab, January 2023-August 2023

Advisor: Matei Ciocarlei

- **Learned Decision Making with Tactile Sensors**
 - Contributed to the design, training, and testing of Deep Reinforcement Learning policies in Isaac Gym

Rutgers Pracsys Lab (Undergraduate), June 2017-August 2021

Advisor: Kostas Bekris

Teaching Assistant (TA) Experience

- Intro to AI (Fall 2023, Spring 2024), Intro to Comp. Rob. (Fall 2024), Data Sci. (Spring 2025)

Internships

Siemens, May 2022-August 2022

- **Autonomous train**

- Collaborated with LSL at Carnegie Mellon University to develop an autonomous train control system using Reinforcement Learning for the Federal Railroad Administration

Publications

P. Meng, B. Wen, J. Li, J. Yu, K. Bekris, “FoundationCalibration: Marker-Free, Zero-shot Robot-Camera Calibration via Foundation Models”, *Under Review*

J. Doerr*, **P. Meng***, S. Tangirala*, K. Bekris, “Integrating Explicit and Implicit Models for Robot Motion Generation,” *ISRR 2026*, Nov. 2026.. *Equal Contribution

N. Chen, **P. Meng**, C. Tang, A. DeGay, Z. Brei, R. Kramer-Bottiglio, K. Bekris, M. Aanjaneya,” Model Predictive Control of Tensegrity Robots via Contact-Aware Graph Neural Dynamics Model” *IROS 2026*, Sep. 2026

E. Granados, **P. Meng**, C. Tang, S. Sangani, W. Johnson, R. Kramer-Bottiglio, K. Bekris, “State and Trajectory Estimation of Tensegrity Robots via Factor Graphs and Chebyshev Polynomials,” *Robosoft 2026*, Apr. 2026

W. Johnson*, **P. Meng***, N. Chen, L. Cimatti, A. Vercoutere, M. Aanjaneya, R. Kramer-Bottiglio, K. Bekris, “[Demonstrating an Open-source, Reproducible System for Tensegrity Navigation among Obstacles](#)” *RA-L*, Mar. 2026. To be presented at IROS 2026, Sept. 2026. *Equal Contribution

J. Xu, Y. Jia, D. Yang, **P. Meng**, X. Zhu, Z. Guo, S. Song, and M. Ciocarlie, "[Tactile-based Object Retrieval From Granular Media](#)," *Autonomous Robots*, 49, Article 32. Aug. 2025
doi:10.1007/s10514-025-10212-9

K. Bekris, J. Doerr, **P. Meng**, S. Tangirala, “[The State of Robot Motion Generation](#),” *ISRR 2024*, Dec. 2024

P. Meng, W. Wang, D. Balkcom, and K. E. Bekris, “[Proof-Of-Concept Designs For the Assembly of Modular Dynamic Tensegrities into Easily Deployable Structures](#),” *Earth and Space 2021*, Sep. 2021. doi:10.26226/m.632b0aa4f30377bc3bafa226

H. Lim, A. Poleksic, Y. Yao, H. Tong, D. He, L. Zhuang, **P. Meng**, and L. Xie, "Large-Scale Off-Target Identification Using Fast and Accurate Dual Regularized One-Class Collaborative Filtering and Its Application to Drug Repurposing," *PLoS Comput Biol*, vol. 12, no. 10, pp. e1005135, Oct. 2016. doi: 10.1371/journal.pcbi.1005135.